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Does $[\text{TF}_2\text{N}]^-$ slither? Equivalence of cation and anion self-diffusion activation volumes in 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide†

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New high-pressure self-diffusion data are reported for the ionic liquid 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide ([EMIM][TF_2N]) at pressures up to 363 MPa in the temperature range 288–348 K. The cation and anion activation volumes derived from these are found to be equal at a fixed temperature, within experimental error, in contradiction to a report in the literature that they differ significantly. Self-diffusion activation volumes derived from our earlier high-pressure diffusion studies also show equality for the respective cations and anions of bis(trifluoromethylsulfonyl)amide, tetrafluoroborate and hexafluorophosphate salts with various cations. Stokes–Einstein–Sutherland analysis and density scaling are applied to the [EMIM][TF_2N] self-diffusion measurements and support the conclusion that pressure effects both cation and anion mass (and hence charge) transport in the same way. The density scaling parameters are consistent with the theoretical predictions of Knudsen *et al.* and agree with that for the viscosity, as for other ionic liquids.

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In previous studies we have reported on the pressure and temperature dependence of ion self-diffusion coefficients,^{1–7} the electrical conductivity^{1,8–10} and the viscosity^{2,6,11–16} of a range of ionic liquids. In statistical mechanical Green–Kubo theory, the self-diffusion coefficients and the conductivity can be written in terms of velocity cross-correlation coefficients (VCC) and it has been found that these show, for a given ionic liquid, very similar viscosity dependence for cation–cation, anion–anion, and cation–anion interactions. For a one component system, the VCC are all negative, corresponding to overall *anti*-correlations of the time and ensemble averages of the ion velocities, a nonintuitive result.¹⁷ Non-equilibrium thermodynamics Laity resistance coefficients have also been examined as an alternative description, but these are more useful in determining ion-association, of which there appear to be very few confirmable instances.^{18,19}

An alternative approach is to use Eyring activation volumes, $\Delta V_i^\ddagger \equiv RT(\partial \ln D_{\text{Si}}/\partial p)_T$, where D_{Si} is the ion self-diffusion coefficient and p is the pressure. Suarez *et al.*^{20,‡} have

reported that diffusion of the anions is slowed much more by increasing the pressure than for the cations in the ionic liquid 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide ([EMIM][TF_2N]), with the anion volume of activation ($\Delta V_a^\ddagger$) being twice that of the cation (for the single isotherm studied). As a consequence, this implies, *inter alia*, different Stokes–Einstein–Sutherland (SES) slopes²¹ for the two ions in correlations with the viscosity, not seen in other high-pressure ionic liquid studies where SES slopes for cation and anion self-diffusion are found to be equal and also equal to that obtained from Walden plots of the molar conductivity for the same ionic liquid.^{1–7}

$$\text{SES: } D_{\text{Si}}/T \propto (1/\eta)^t; \quad \text{Walden: } \Lambda \propto (1/\eta)^t \quad (1)$$

Suarez *et al.*²⁰ did not use SES analysis as high-pressure viscosities of [EMIM][TF_2N] were then unavailable in the literature, though these have since been determined in our laboratory.¹⁶

Table 1 shows examples of the similarity of the cation and anion volumes of activation calculated from the experimental data for 1-butyl-3-methylimidazolium and 1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amides. The value reported by Suarez *et al.* for $\Delta V_a^\ddagger([\text{TF}_2\text{N}]^-)$ in [EMIM][TF_2N] at room temperature (~ 295 K), $29 \text{ cm}^3 \text{ mol}^{-1}$, is similar in magnitude. It is the lower value reported for $[\text{EMIM}]^+$, $14.6 \text{ cm}^3 \text{ mol}^{-1}$, that appears to be anomalous in this group of ionic liquids.

Suarez *et al.*²⁰ suggested that the translational mobility of the anion is facilitated by interconversion between *cis* and *trans* conformers, so that the $[\text{TF}_2\text{N}]^-$ ion in [EMIM][TF_2N], “ambulates through an ionic liquid by twisting between its two conformers,

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‡ Suarez *et al.* (ref. 20) use the common acronym [TFSA][−] to represent the bis(trifluoromethylsulfonyl)amide ion. [TFSI][−] is another common variant. We use the abbreviation $[\text{TF}_2\text{N}]^-$ here for consistency with our earlier publications. Unfortunately, there is no agreed abbreviation system for the complex ion structures found in ionic liquids.

Table 1 Activation volumes ($\Delta V^\ddagger$) and SES and Walden relation exponents (t) for ionic liquid self-diffusion coefficients at 0.1 MPa^a

IL ^b	T/K	$\Delta V^\ddagger$ cation	$\Delta V^\ddagger$ anion	t cation	t anion	t cond.
[EMIM][Tf ₂ N] ²⁰	~295	14.6	28.8	0.62 ^b	1.2 ^b	—
[BMIM][Tf ₂ N] ⁴	298.15	28.1	25.4	0.92 ₆	0.93 ₆	0.93 ₀
	323.15	22.5	23.4			
[HMIM][Tf ₂ N] ⁷	323.15	25.2	24.6	0.94 ₂	0.94 ₈	0.94 ₁

^a $\Delta V^\ddagger$ units: cm³ mol⁻¹. Calculated from data in the references cited. The $\ln D_{\text{Si}}$ isotherms are assumed linear in p for this purpose, though in fact the longer isotherms obtainable at higher temperatures are usually non-linear, as is common for liquid transport properties, making $\Delta V^\ddagger$ pressure dependent with $\partial(\Delta V^\ddagger)/\partial p_T < 0$ below 300–400 MPa (see below). ^b Calculated using viscosities from ref. 16.

similar to how a snake moves by slithering.” The difference between the two volumes of activation led to the deduction that this “slithering” process is hindered by the reduction in free volume resulting from the application of high pressure. If true, this is a remarkable mechanism for diffusion given the apparent absence of any such effect in [BMIM][Tf₂N] and [HMIM][Tf₂N] (Table 1).

Differences in the molar volumes of the *cis* and *trans* [Tf₂N]⁻ isomers estimated from very high-pressure infrared and Raman spectroscopy are quite small, about 0.3–0.4 cm³ mol⁻¹ in some reports^{22,23} and 0.7 cm³ mol⁻¹ in others,²⁴ though there is not necessarily a direct link between molar and activation volumes. Yoshimura *et al.*²⁵ have emphasized that for 1-alkyl-3-methylimidazolium salts, pressure-induced changes in isomeric population ratios are quite small and remarked that “the liquid structure of [C_nmim][TFSI] tends to resist shear stress through cooperative conformational changes in the cation and anion.”²⁵

The new self-diffusion measurements on [EMIM][Tf₂N] provided here give a more extensive data set for the examination of ion mass transport in this liquid, extending to higher pressures and temperatures. Volumes of activation are determined and compared with those for [BMIM][Tf₂N], [HMIM][Tf₂N] and other ionic liquids. The consistency of our results is examined with the SES equation and density scaling.

Experimental

[EMIM][Tf₂N] was obtained from Solvionic SA, France, for the high-pressure measurements. Aliquots were transferred to Teflon/Nylon high-pressure bellows cells inside a dry glove box just prior to use. The water content of the sample was determined by Karl–Fischer titration. The sample characteristics are given in Table 2. A different sample was used for ‘atmospheric’ pressure check measurements, a dried, out-gassed sample prepared as in earlier ionic liquid studies,²⁶ sealed *in vacuo* directly into a 5 mm NMR tube. It was described and used in earlier work.⁴

The steady-gradient spin-echo NMR self-diffusion apparatus, based on a Bruker Minispec PC-100 unit, and its calibration and application to ionic liquids have been described in previous work.^{1,2,4,5} As is usual for ionic liquids, the temperature range

that could be covered by our apparatus was limited by the short T_2 relaxation times, which attenuate the signal in addition to diffusion, with greater effect at lower temperatures, giving low signal to noise ratios. The high-pressure isotherms were normalised on the atmospheric pressure results (obtained with a separate dedicated probe with the same combined rf/gradient coil unit design²⁷ as the pressure vessel) to remove any calibration difference between the low and high-pressure magnetic field gradient coils (the latter is contained within the pressure vessel). Atmospheric pressure values have been published previously.⁴ The temperature was maintained by circulating water or ethanol (for low temperatures) from a Julabo circulating thermostat (FP40-JC or FP50-JC) through a coil wrapped around the Be–Cu alloy pressure vessel, both contained within a large glass Dewar set between the poles of the NMR electromagnet. It was determined with a calibrated Pt-thermometer element set in the bottom closure of the pressure vessel, the resistance of which was measured with a Keithley 197 digital multi-meter (4-wire method). Pressures, produced with a Pressure Products Industries 400 MPa hand-pump, were measured with a Heise Bourdon 500 MPa gauge (Dresser Instruments, model CM). For proton measurements, 3 M Fluorinert FC-75 electronic liquid was used as the hydraulic fluid and for fluorine measurements diethyl sebacate was employed. The estimated relative standard uncertainty in the self-diffusion coefficients D_{Si} is $u_r = 3\%$ and the temperature and pressure uncertainties $u(T) = 0.01$ K and $u(p) = 0.2$ MPa above 0.1 MPa, respectively.

Results and discussion

Self-diffusion coefficients

Self-diffusion coefficients (D_{Si}) for [EMIM][Tf₂N] are given in Tables S1 and S2 of the ESI.† Densities, calculated from the equations of state employed for high-pressure viscosity determinations,¹⁶ are also given as they are used for the density scaling of D_{Si} .

We have previously published self-diffusion coefficients for this ionic liquid at atmospheric pressure.⁴ Table 3 gives the coefficients of best fit for the atmospheric pressure ion self-diffusion coefficients for the Litovitz (2) and VFT (3) equations for both our data set and that of Saurez *et al.*

$$\ln D_{\text{Si}} = A + B/T^3 \quad (2)$$

$$\ln D_{\text{Si}} = A' + B'/(T - T_0) \quad (3)$$

Table 2 Sample description

Sample	M^a / g·mol ⁻¹	CAS no.	Source	Purity ^b / mol%	10^6 $w(\text{H}_2\text{O})^{ac}$
A	391.311	174899-82-2	Our synthesis ^{4,26}	> 99	305
B			Solvionic SA, France	> 99.5 ^d	340

^a M = molar mass; w = weight fraction. ^b Based on ¹H NMR analysis. ^c Karl Fischer coulometric water analysis. ^d Manufacturer's analyses: lot no. L 16 031 401, $10^6 w(\text{H}_2\text{O}) = 150$, [Li] = 0.002%, [methylimidazole] = 0.0012%.

Table 3 Coefficients of best fit to the Litovitz (1) and VFT (2) equations for the self-diffusion coefficients, D_{Si}^a

	[EMIM][Tf ₂ N]		[EMIM][Tf ₂ N] ^b	
	$D_{S+}^a/10^{-12} \text{ m}^2 \text{ s}^{-1}$	$D_{S-}^a/10^{-12} \text{ m}^2 \text{ s}^{-1}$	$D_{S+}^a/10^{-12} \text{ m}^2 \text{ s}^{-1}$	$D_{S-}^a/10^{-12} \text{ m}^2 \text{ s}^{-1}$
Litovitz, eqn (1)				
A	7.6427 ± 0.021	7.2746 ± 0.018	8.100 ± 0.20	6.683 ± 0.10
$B \times 10^{-6}/\text{K}^3$	-97.146 ± 0.58	-101.965 ± 0.56	-120.67 ± 7.3	-73.27 ± 3.5
T range/K	268–362	278–363	296–373	298–373
Relative standard uncertainty of fit, $100u_r$	3.1 ^c	2.0	17	7
VFT, eqn (2)				
A'	10.173 ± 0.16	9.218 ± 0.19	—	—
B'/K	-1068.2 ± 59	-879.0 ± 63	—	—
T_0/K	126.34 ± 4.9	146.36 ± 5.8	—	—
Relative standard uncertainty of fit, $100u_r$	2.1	1.8	—	—

^a An additional, nonsignificant digit is given for each fitted parameter to reduce the effects of rounding errors on any calculations employing the regressions. ^b Calculated from data in ref. 20. ^c The Litovitz equation fits less well here at the temperature extremes due to the extended temperature range. The VFT equation is better in this case.

Suarez *et al.*²⁰ determined self-diffusion coefficients for [EMIM][Tf₂N] and analogues with deuterium isotopologues of the cation at atmospheric pressure in a variable temperature probe using pulsed gradient methods. The deuterium-labelled cations were 3-ethyl-1-(methyl-*d*₃)-imidazolium (CAS No. 769874-37-5, [EMIM-MD₃]⁺ in Saurez' notation), 3-(ethyl-2,2,2-*d*₃)-1-methylimidazolium (CAS No. 1287287-74-4, [EMIM-TD₃]⁺), and 3-(ethyl-1,1-*d*₂)-1-methylimidazolium (CAS No. 1287287-76-6,

[EMIM-D₂]⁺). Fig. 1 shows the deviations from our data, as fitted to eqn (3).

It is clear from the figure that there is a great deal of scatter in individual data sets, ranging from 7 to 17% for the cations and from 5 to 8% for the anion, based on the standard deviations of fits to eqn (1). Fits to eqn (2), with an additional parameter, are not meaningful with such scatter. Some data sets show larger scatter than Saurez' estimate of a precision of 5%. The deviations from our results are larger still.

Suarez *et al.* attempted to explain the differences between the self-diffusion coefficients of the isotopologues, particularly at the higher temperatures, in terms of "three competing factors: the mass difference between ¹H and ²H, and the strengths of the electric field gradients, and the reduction in hydrogen bonding with increasing temperatures." However, such very large isotope effects as reported [as big as 1.69 for $D_s([EMIM]^+)/D_s([EMIM-D_2]^+)$ at 373 K, (see Fig. 5 of ref. 20)] are inconsistent with what is known from radio-tracer and NMR studies.^{28,29} These are most marked when the isotopologues are both strongly H-bonded and substitution changes the moment of inertia of the diffusing species: thus the ratios $D_s(\text{H}_2\text{O})/D_s(\text{D}_2\text{O})$ and $D_s(\text{NH}_3)/D_s(\text{ND}_3)$ at 298 K are 1.23 and 1.15 respectively.^{30,31} Smaller ratios apply when there is no H-bonding, e.g. $D_s[(\text{CH}_3)_2\text{SO}]/D_s[(\text{CD}_3)_2\text{SO}]$ and $D_s(\text{C}_6\text{H}_6)/D_s(\text{C}_6\text{D}_6)$ at 298 K are 1.12 and 1.06 respectively, the latter mainly due to the difference in mass, consistent with similar ratios for the fluidities.²⁹ On the basis of the moment-of-inertia findings of Holz *et al.*,²⁹ any isotope effect should be largest for [EMIM-TD₃][Tf₂N], relative to the undeuteriated form, not [EMIM-D₂][Tf₂N]. Despite the speculation by Saurez *et al.*²⁰ about the effects of H-bonding between cations and anions on the transport properties of the ions in [EMIM][Tf₂N], we would argue that the fact that the Laity resistance coefficients are positive, as found in our previous study at atmospheric pressure⁴ indicates that there is no cation-anion association in this substance.¹⁸

The high-pressure self-diffusion data, plotted as functions of temperature and pressure for [EMIM][Tf₂N], together with the Saurez data set obtained with a high-pressure vessel without any temperature control, are shown in Fig. 2. The disparity between the pressure dependence of their results and ours is very clear.

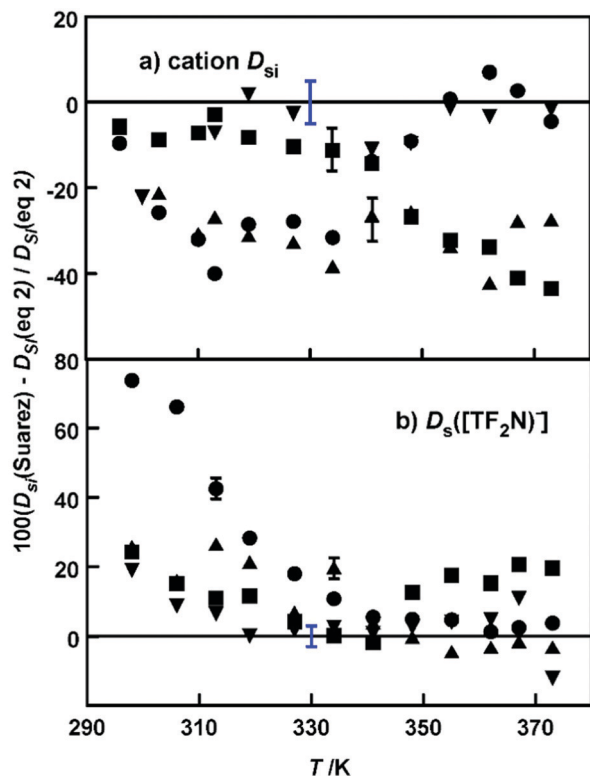


Fig. 1 Deviations of the ion self-diffusion atmospheric pressure data of Suarez *et al.*²⁰ for [EMIM][Tf₂N] from the results of this work, fitted to eqn (2). (a) Cation isotopologues. (b) [Tf₂N][−]. Symbols: circles, [EMIM][Tf₂N]; squares, [EMIM-D₂][Tf₂N]; triangles, [EMIM-TD₃][Tf₂N]; inverted triangles, [EMIM-MD₃][Tf₂N]. Error bars: blue, this work, ±3%; black, Suarez *et al.*, ±5%.

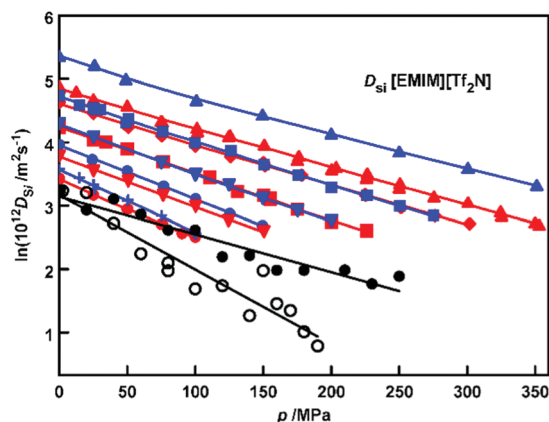


Fig. 2 Pressure dependence of the self-diffusion coefficients (D_{Si}) of [EMIM][Tf₂N]: this work, blue symbols, cation, $D_{\text{S}+}$; red symbols, anion, $D_{\text{S}-}$, (note that $D_{\text{S}+} > D_{\text{S}-}$); crosses, 288 K (cation only); circles, 298 K, inverted triangles, 308 K; squares, 323 K; diamonds, 338 K (anion only); triangles, 343 K; Saurez *et al.*,²⁰ 295 K, black symbols; solid circles, $D_{\text{S}+}$; open circles, $D_{\text{S}-}$.

As in our previous high-pressure studies, modified Litovitz (ML) and VTF (MVFT1, MVFT2) equations have been used to fit the complete data sets given in the ESI:[†]

$$D_{\text{Si}} = \exp(a + bp + (c + dp + ep^2)/T^3) \text{ (ML)} \quad (4)$$

$$D_{\text{Si}} = \exp(a' + b'p + (c' + d'p + e'p^2)/(T - T_0)) \text{ (MVFT1)} \quad (5)$$

and

$$D_{\text{Si}} = \exp(a'' + b''p + \delta T_0(p)/(T - T_0)),$$

$$T_0(p) = c'' + d''p + e''p^2 \text{ (MVFT2)} \quad (6)$$

The MVFT1 equation has a pressure-dependent Angell strength factor [$\delta = (c' + d'p + e'p^2)/T_0$], whereas the MVFT2 form has a pressure-dependent T_0 . The fitted coefficients and relative standard uncertainty of fit, u_r , are given in Table 4. Deviation plots for the MVFT1 equation are given as Fig. S1 and S2 in the ESI.[†]

Empirical viscosity relationships

Fig. 3 shows Stokes–Einstein–Sutherland (SES) for the self-diffusion coefficients. Note that $\ln(D_{\text{Si}}/T)$ is inversely proportional to the friction coefficient in the original SES model, $(D_{\text{Si}}/k_{\text{B}}T)$ being the diffusive mobility.^{2,21} As in our earlier high-pressure ionic liquid studies, the ion isotherms overlap one another and the atmospheric pressure isobar, with very similar exponents, t , for both self-diffusion coefficients and the atmospheric pressure molar conductivity (not plotted),^{4,32} consistent with all three quantities being functionals of either velocity auto- or cross-correlation coefficients (Table 5).^{5,33} The corresponding plots for the Suarez data for [EMIM][Tf₂N] are also shown in the upper panel of Fig. 3. The cation values are too low, the atmospheric pressure self-diffusion coefficient being half that of this work and 15% below that obtained with the Suarez atmospheric pressure probe. The SES exponent, t , is very low, 0.62 ± 0.05 , much smaller than we have found in any of our ionic liquid

Table 4 Coefficients of best fit for the modified Litovitz equation, eqn (4), and the modified VFT equations, eqn (5) and (6), for the self-diffusion coefficients of [EMIM][Tf₂N] for the temperature and pressure ranges indicated^a

	$10^{12} D_{\text{S}+}/\text{m}^2 \text{ s}^{-1}$	$10^{12} D_{\text{S}-}/\text{m}^2 \text{ s}^{-1}$
<i>T</i> range/K	268–362	278–363
Maximum pressure, $p_{\text{max}}/\text{MPa}$	351	353
ML, eqn (4)		
a	7.6273 ± 0.015	7.2833 ± 0.016
$b \times 10^3/\text{MPa}^{-1}$	-2.076 ± 0.28	-1.531 ± 0.24
$c \times 10^{-6}/\text{K}^3$	-96.897 ± 0.42	-102.336 ± 0.52
$d \times 10^{-6}/(\text{K}^3 \text{ MPa}^{-1})$	-0.1917 ± 0.007	-0.2015 ± 0.007
$e/(\text{K}^3 \text{ MPa}^{-2})$	107.9 ± 19	28.0 ± 13
Relative standard uncertainty of fit, $100u_r$	2.8	2.4
MVFT1, eqn (5)		
a'	10.082 ± 0.13	9.010 ± 0.17
$b' \times 10^3/\text{MPa}^{-1}$	1.047 ± 0.39	1.737 ± 0.40
c'/K	-1038.0 ± 46	-807.0 ± 56
$d'/(\text{K MPa}^{-1})$	-1.7042 ± 0.076	-1.5619 ± 0.094
$e' \times 10^3/(\text{K MPa}^{-2})$	0.6019 ± 0.083	0.1212 ± 0.061
T_0/K	128.72 ± 3.9	153.64 ± 5.5
Relative standard uncertainty of fit, $100u_r$	2.1	2.4
MVFT2, eqn (6)		
a''	10.016 ± 0.13	9.004 ± 0.17
$b'' \times 10^3/\text{MPa}^{-1}$	-1.610 ± 0.26	-1.649 ± 0.28
δ	-7.769 ± 0.57	-5.191 ± 0.52
c''/K	130.66 ± 3.8	154.52 ± 5.2
$d'' \times 10^3/(\text{K MPa}^{-1})$	87.95 ± 3.5	96.62 ± 4.1
$e'' \times 10^6/(\text{K MPa}^{-2})$	-66.46 ± 5.1	-51.8 ± 11
Relative standard uncertainty of fit, $100u_r$	2.2	2.4

^a An additional, nonsignificant digit is given for each fitted parameter to reduce the effects of rounding errors on any calculations employing the regressions.

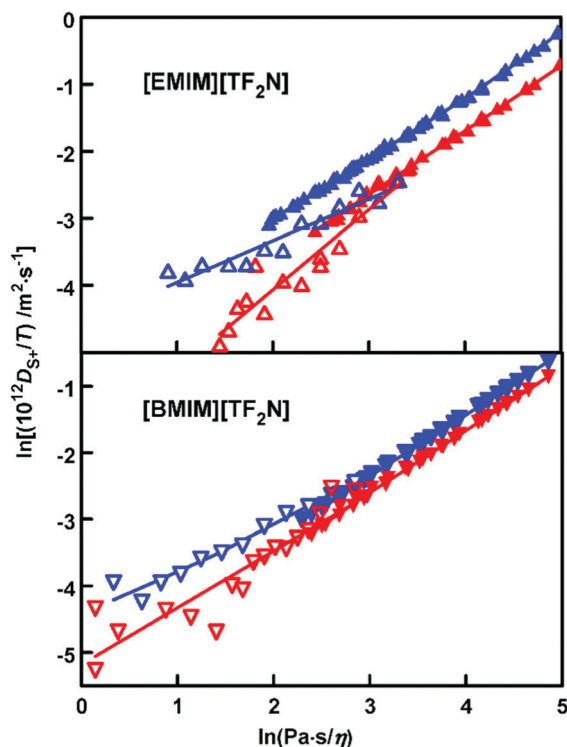


Fig. 3 Stokes–Einstein–Sutherland of self-diffusion coefficients, D_{Si} , for [EMIM][Tf₂N] [upper panel] and [BMIM][Tf₂N] [lower] panel against reciprocal viscosity ($1/\eta$). Symbols: blue, cations; red, anions; filled symbols, this work; open triangles, Saurez *et al.*²⁰

studies. For the anion, the atmospheric pressure self-diffusion coefficient is in good agreement with our value, but the SES exponent, t , is now too large, 1.2 ± 0.1 , an unphysical value as t should not be greater than unity.²¹ There is no agreement with the atmospheric pressure data, where the cation exponent is also unphysical, 1.2, and that for the anion is too low, 0.62, due to the high scatter shown in Fig. 1b. Isotherms and isobars normally overlap in ionic and molecular liquid SES plots with the only known exceptions being water and alcohols, due to their strongly-bound, open H-bonded liquid structures.^{1–6,34}

The Suarez group have also published self-diffusion data for [BMIM][Tf₂N], again at approximately 295 K, (D_{S+} : 230 MPa maximum; D_{S-} : 250 MPa maximum). Here the derived $\Delta V^\ddagger$ values are similar, 31.7 and 26.7 cm³ mol^{−1}, for the cation (averaged over [BMIM]⁺ and two *D*-substituted isotopologues) and anion respectively. These are more in accord with the values in Table 1 derived from our previously published data set.⁴ On the other hand, Suarez *et al.* found low values for $\Delta V^\ddagger$ for both ions in each of the tetrafluoroborate and bis(fluoromethylsulfonyl)imide salts, [EMIM]BF₄ (14 and 15 cm³ mol^{−1}) and [EMIM][FSI] (13 and 12 cm³ mol^{−1}), also at 295 K.[§]

§ The high-pressure NMR diffusion apparatus was decommissioned in June 2021 as it was becoming increasingly difficult to maintain the magnet power supply and system electronics. Hence we did not attempt to study [EMIM]BF₄ or [EMIM][FSI].

Table 5 Exponents (t) for Stokes–Einstein–Sutherland and Walden fits of the self-diffusion coefficients, D_{Si} , molar conductivity Λ and distinct diffusion coefficients, D_{ij}^d and the consistency test plot of $\ln(\Lambda T)$ against $\ln(D_{S+} + D_{S-})$ for [EMIM][Tf₂N]

$t(D_{S+})$	$t(D_{S-})$	$t(\Lambda)$	$\ln(\Lambda T)$ vs. $\ln(D_{S+} + D_{S-})$	$t(D_{++}^d)$	$t(D_{--}^d)$	$t(D_{+-}^d)$
0.91 ₆	0.95 ₇	0.89 ₉ ^a	0.95 ₄ ^a	0.96 ₀ ^a	0.98 ₅ ^a	0.89 ₉ ^a

^a Atmospheric pressure values from ref. 4.

For comparison with [EMIM][Tf₂N], SES plots for [BMIM][Tf₂N] are given in the lower panel of Fig. 3. Though it is difficult to be certain due to the large scatter at high viscosity, at first sight the slight upturn in the [BMIM]⁺ isotherm appears physically reasonable as decoupling of self-diffusion and viscosity is known to occur in fragile liquids at high viscosities,³⁵ though these are usually produced by lowering the temperature to near glass-transition levels. The glass transition temperature (T_g) at 250 MPa, estimated using Angell's relationship between T_g and $T_0(p)$, derived from an MVFT2 fit to the high pressure viscosity¹³ [the analogue of eqn (5)], is only 205 K. Consequently, the upturn in the Suarez isotherm at 295 K is unlikely to be due to decoupling as this usually occurs only below $1.2T_g$,³⁵ 246 K in this case. Given the trends shown by the [EMIM][Tf₂N] data, the apparent upturn probably reflects the greater experimental uncertainty in the measurement of very low self-diffusion coefficients.

Thermodynamic scaling

Thermodynamic (or density) scaling has been applied to the self-diffusion and the conductivity data sets, as in our earlier work.^{4,6,36,37} This method expresses liquid transport properties as functions of the group (TV^γ) where the scaling constant γ is the fitted parameter. It can be related to the steepness of the repulsive part of the intermolecular potential for model fluids but is moderated by attractive forces in real fluids.^{38,39} It has been found that dimensionless Rosenfeld reduced transport properties should be used for thermodynamic scaling.⁴⁰ For molecular liquids, this yields very similar (but unequal) scaling parameters for high-pressure experimental self-diffusion coefficients and viscosities,³⁶ as well as thermal conductivities.³⁷ The latter property does show some variability, primarily for larger, polyatomic molecules where internal modes may well play a significant part in energy transport, and also for the Lennard-Jones fluid, which may be due to a different dependence on packing fraction to the other transport properties.

For the viscosity, η , self-diffusion coefficient, D_{Si} , conductivity and molar conductivity the appropriate expressions are:

$$\tilde{\eta} = \eta v^{2/3} / \sqrt{mk_B T} \quad (7)$$

$$\tilde{D}_{Si} = D_{Si} v^{-1/3} \sqrt{m/k_B T} \quad (8)$$

$$\tilde{\kappa} = \kappa \sqrt{mk_B T} / [e^2 v^{-2/3}] \quad (9)$$

$$\tilde{\Lambda} = \Lambda \sqrt{mk_B T} / [N_A e^2 v^{1/3}] \quad (10)$$

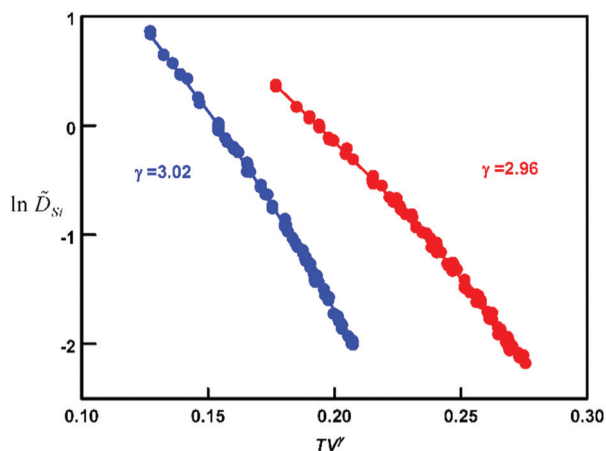


Fig. 4 Density scaling plot of the reduced self-diffusion coefficients of [EMIM][Tf₂N] against $(TV)^\gamma$. Units: T/K , $V/\text{cm}^3 \text{ mol}^{-1}$; $\tilde{D}_{Si}$ is dimensionless. Symbols: blue, cation; red, anion.

where v is the molecular volume (V_m/N_A), m is the molecular mass (M/N_A), k_B is Boltzmann's constant, and e is the electronic charge.[¶]

As in ref. 4 and our other work,^{15,16,37} the logarithms of the Rosenfeld reduced quantities were fitted to polynomials in $1/(TV)^\gamma$ for selected values of γ with polynomial orders from 2 to 3, depending on the range of available data, with the order being kept as low as needed to give sensible fits commensurate with the experimental precision; γ is chosen by iteration to minimize the standard deviation of the fit. Fig. 4 shows the density scaling plot for the self-diffusion coefficients. Deviation plots for the scaling are given as Fig. S3 and S4 in the ESI.[†]

In the case of ionic liquids, the γ parameter is almost identical for the transport properties (viscosity, ion self-diffusion coefficients and electrical conductivity).^{4,6,16} Such high-pressure thermal conductivity sets as there are in the literature do not extend to high enough pressures to permit a test for that property. As is usual, the scaling parameter for the experimental self-diffusion coefficients are less than that for the reduced self-diffusion coefficients, whereas that for the experimental viscosity is more than that for the reduced viscosity. The scaling parameters converge on using the reduced transport properties (Table 6), consistent with earlier results for ionic liquids.⁴

Knudsen *et al.*⁴¹ have conducted simulations of a model fluid with a repulsive inverse power law (IPL) part and an attractive coulombic part to the intermolecular potential of point particles. Partial structure factors and transport properties were simulated over broad ranges of temperature and density.

They demonstrated that the model has lines in its phase diagram where the dynamical and transport (D_s , η) properties are invariant, consistent with the constancy of γ found in experiment, and that the low values of γ displayed by ionic liquids relative to uncharged molecular liquids are due to the

Table 6 Thermodynamic scaling parameters γ for the ion self-diffusion coefficients (D_{Si}) and viscosities (η) and the corresponding reduced transport properties ($\tilde{D}_{Si}$, $\tilde{\eta}$) for [EMIM][Tf₂N]^a

Scaling parameter ^{ab}		$u_t(\text{fit})^c/\%$
$\gamma(D_{S+})$	2.87	2.5
$\gamma(\tilde{D}_{S+})$	3.02	2.4
$\gamma(D_{S-})$	2.81	3.1
$\gamma(\tilde{D}_{S-})$	2.96	2.9
$\gamma(\eta)$	3.23	2.0 ¹⁶
$\gamma(\tilde{\eta})$	2.97	2.7 ¹⁶

^a Boldface numbers denote scaling parameters for the reduced properties. ^b The uncertainty, u , in γ is 0.03. ^c $u_t(\text{expt})/\%$ = 3 for each diffusion coefficient data set.

coulombic attractions of oppositely charged ions. Isomorph theory⁴² can be used to express the scaling parameter γ in terms of fluctuations in the potential energy, U , and the thermodynamic virial, $W = (pV - RT)$, from their thermodynamic averages. Knudsen *et al.*⁴⁰ showed γ can be approximated by

$$\gamma = (3 + X/3)/(1 + X) \quad (11)$$

where X is the ratio of the variance of Coulomb potential energy fluctuations, $\langle(\Delta U_C)^2\rangle$, to that for IPL energy fluctuations, $\langle(\Delta U_{\text{IPL}})^2\rangle$, it being assumed the two are uncorrelated. For an IPL fluid with an intermolecular potential varying with the separation r as r^{-n} , γ is $n/3$,^{43,44} so the limit of 3 at $X = 0$ in eqn (11) for a molecular fluid with no coulombic interactions comes from the particular choice of $n = 9$ for the simulations of Knudsen *et al.*

Values of γ displayed by ionic liquids, as shown in Table 6, therefore, are consistent with moderation of the effects of the repulsive van der Waals interactions between the ions which lead to larger γ by the coulombic attractions of oppositely charged ions that reduce it. The similarity of the γ values for the two reduced ion self-diffusion coefficients is further evidence for the argument that pressure effects both cation and anion mass (and hence charge) transport in the same way.

[EMIM][Tf₂N]: "Does TFSA Slither?"

The major conclusion of the self-diffusion study of Suarez *et al.*²⁰ was: "Most significantly, the pressure dependences of the EMIM and TFSA self-diffusion coefficients revealed that the displacements of the cations and anions are independent, with diffusion of the TFSA anions being slowed much more by increasing pressure than for the EMIM cations, as shown by their respective activation volumes ($28.8 \pm 2.5 \text{ cm}^3 \text{ mol}^{-1}$ for TFSA versus $14.6 \pm 1.3 \text{ cm}^3 \text{ mol}^{-1}$ for EMIM). Increasing pressure may lower the mobility of the TFSA anion by hindering its interconversion between *trans* and *cis* conformers, a process that is coupled to diffusion according to published molecular dynamics simulations."

From the evidence examined above it is clear that (1) our data do not support Suarez' claims: the volumes of activation for the anion and cation self-diffusion coefficients do not differ significantly, as shown in Fig. 5 and Table 7, which includes activation volumes derived for all the ionic liquids in our high

[¶] There are typographical errors in eqn (15) and (16) of ref. 4, repeated in ref. 6 and 37. Eqn (9) and (10) of this work are the correct versions. The scaling parameters given in Table 8 of ref. 4 and Table 4 of ref. 37 were calculated correctly.

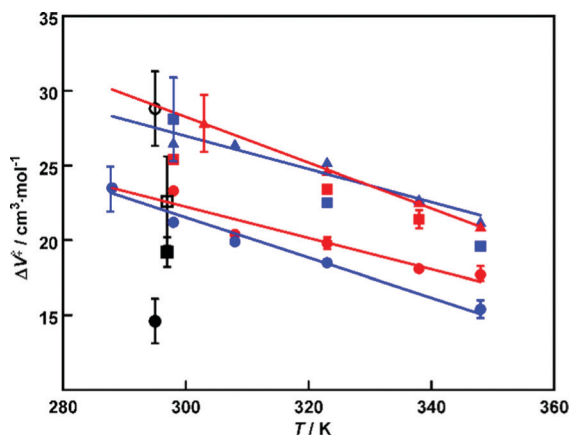


Fig. 5 Volumes of activation, $[\Delta V^\ddagger = RT(\partial \ln D_{\text{Si}}/\partial p)_T]$, for $[\text{Tf}_2\text{N}]^-$ -ionic liquids derived from linear fits of $\ln D_{\text{Si}}$ as a function of pressure. Symbols: blue, cations; red, anions; this work, solid circles, $[\text{EMIM}][\text{Tf}_2\text{N}]$; squares, $[\text{BMIM}][\text{Tf}_2\text{N}]$ (derived from the data of ref. 4); triangles, $[\text{HMIM}][\text{Tf}_2\text{N}]$ (derived from the data of ref. 7); black, Suarez *et al.*,²⁰ circles, $[\text{EMIM}][\text{Tf}_2\text{N}]$; squares, $[\text{BMIM}][\text{Tf}_2\text{N}]$; solid, cations; open, anions. The upper lines combine the $[\text{BMIM}][\text{Tf}_2\text{N}]$ and $[\text{HMIM}][\text{Tf}_2\text{N}]$ results from this laboratory^{4,7} to simplify the figure. The error bars combine the uncertainty of the measurements ($\pm 3\%$ in D_{Si}) with that of the linear fit for each isotherm, which is larger for the shorter, lower temperature isotherms (see Table 7). The apparent difference between cation and anion at higher temperatures for $[\text{EMIM}][\text{Tf}_2\text{N}]$ is due in part to the linearisation of curved isotherms (ρ^2 terms are required in eqn (4)–(6)). The pressure dependence of $\Delta V^\ddagger$, derived from eqn (5), is shown in Fig. 6.

pressure works: the recent molecular dynamics simulations of Kubisiak *et al.*⁴⁵ are consistent with this result; (2) the pressure dependence of the D_{Si} is similar to that for $[\text{BMIM}][\text{Tf}_2\text{N}]$ and $[\text{HMIM}][\text{Tf}_2\text{N}]$; (3) all three ionic liquids yield typical Stokes–Einstein–Sutherland diffusion–viscosity plots with overlaid isotherms and isobars collapsed onto a single line and with fractional exponents between 0.9 and 1, similar to other ionic and non-associated molecular liquids, whereas the Suarez data are inconsistent, with different slopes for the high-pressure isotherms and atmospheric pressure isobars; (4) the present data conform to thermodynamic or density scaling, with very similar scaling constants for the self-diffusion coefficients and the viscosity, again consistent with results for other ionic liquids. So the conclusion must be that there is nothing significantly different in the effect of pressure on the self-diffusion of the ions in $[\text{EMIM}][\text{Tf}_2\text{N}]$ to support a different “slithering” mechanism for the $[\text{Tf}_2\text{N}]^-$ ($[\text{TFSA}]^-$) ion in this ionic liquid distinct from that of other cations or anions or in $[\text{Tf}_2\text{N}]^-$ salts with larger 1-alkyl-3-methylimidazolium cations.

For completeness, Fig. 6 shows the pressure dependence of $\Delta V^\ddagger$ for the two self-diffusion coefficients and the fluidity (viscosity data from ref. 10) calculated by differentiating the MVFT1 eqn (5), at 298 and 348 K. These values are biased by the higher temperature isotherms due to the distribution of experimental points, which are fewer at lower temperatures due to the experimental limitations of unfavourable relaxation times, the smallness of the D_{Si} and the correspondingly higher viscosities. The $\Delta V^\ddagger$ decrease with increasing pressure and increasing

temperature. Within the errors derived from the fits, the self-diffusion $\Delta V^\ddagger$ are the same for the cation and anion.

Why then might the Suarez data sets be in error? The scatter in the data is quite high as noted above and the high-pressure values too high for the $[\text{EMIM}]^+$ cation and too low for the $[\text{Tf}_2\text{N}]^-$ anion relative to the atmospheric pressure points. The measurement of self-diffusion coefficients in highly viscous ionic liquids is not straightforward. T_2 relaxation times change with increasing pressure, reducing signal strength, and below around $15 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$, pulsed gradient methods are usually needed, which are inherently more difficult than the steady gradient technique we use.⁴⁶ Suarez *et al.* used both methods, pulsed gradients at atmospheric pressure but steady gradients at high pressure.^{20,47}

For steady gradients, employing the standard Hahn 90–180° NMR rf pulse sequence, the working equation is:⁴⁸

$$A = A_0 \exp\left(-\frac{2}{3}\gamma^2 D_{\text{Si}} \tau^3 [g^2 + 2g \cdot g_0] - \frac{2\tau}{T_2}\right) \quad (12)$$

where A is the measured echo height, γ the gyro-magnetic ratio for the nucleus observed, τ is the rf pulse interval, g the applied magnetic field gradient, g_0 the background gradient, negligible for high-field superconducting magnets but not lower field electromagnets, and T_2 the spin–spin relaxation time. For the high-pressure measurements, Suarez *et al.* used a fixed value of the gradient g obtained by offsetting the probe from the centre of the electromagnet to a position where the magnetic field was not homogeneous as described in more detail in a subsequent paper.⁴⁷ This means that quite large 90–180° radio-frequency pulse intervals were required to attenuate the NMR echo for the small diffusion coefficients measured (see Fig. 3). The signal to noise ratio would also be lower the higher the pressure so the experimental uncertainty can be expected to increase with increasing pressure, in accord with the scatter seen in Fig. 3.

They do not say whether T_2 attenuation corrections were necessary (though ref. 47 implies these were included through the inclusion of the T_2 term in eqn (20): such corrections are more important for the cations where T_2 is shorter than for the anions). They did not employ the cross term, though the offsetting of the probe from the centre of the magnet might well require it. In our work, A_0 , D_{Si} and g_0 (which is very small relative to g and generally reproducible from one experiment to another) are all determined in the least squares regression in variable g experiments at constant τ , with the T_2 term then embedded in the constant A_0 . The results of both positive and negative gradients are averaged. Occasional check experiments with constant g and variable τ , averaged for positive and negative gradients to remove the effect of g_0 , gave results consistent with the variable g experiments where T_2 was sufficiently large, as for the anions in $[\text{EMIM}][\text{Tf}_2\text{N}]$.

For PGSE experiments, the Stejskal–Tanner working equation employed by Suarez *et al.* is

$$A = A_0 \exp[-\gamma^2 g^2 \delta^2 D_{\text{Si}} (\Delta - \delta/3) - 2\tau/T_2] \quad (13)$$

where δ is the width of the gradient pulse (assumed to be rectangular) and Δ is the time interval between the gradient

Table 7 Activation volumes [$\Delta V^\ddagger = RT(\partial \ln D_{\text{Si}}/\partial p)_T$] for 1-alkyl-3-methylimidazolium salts and other ionic liquids, based on linear fits to isotherms^a

[EMIM][Tf ₂ N]			[BMIM][Tf ₂ N] (data: ref. 4)		[HMIM][Tf ₂ N] (data: ref. 7)	
T/K	Cation	Anion	Cation	Anion	Cation	Anion
288	23.5 ± 1.4 ^{ab}	—	—	—	—	—
298	21.2 ± 0.8	23.3 ± 1.0	28.1 ± 2.8	25.4 ± 4.4	26.5 ± 5.3	—
303	—	—	—	—	—	27.8 ± 1.9
308	19.9 ± 0.6	20.4 ± 0.8	—	—	26.4 ± 2.5	—
323	18.5 ± 0.5	19.8 ± 0.7	22.5 ± 0.6	23.4 ± 1.2	25.2 ± 0.9	24.6 ± 0.8
338	—	18.1 ± 0.4	—	21.4 ± 0.6	22.7 ± 0.7	22.5 ± 0.7
348	15.4 ± 0.6 ^c	17.7 ± 0.4	19.6 ± 0.4	19.6 ± 0.6	21.2 ± 0.6	20.9 ± 0.5
[BMIM]BF ₄ (data: ref. 5)			[HMIM]BF ₄ (data: ref. 7)		[OMIM]BF ₄ (data: ref. 5)	
298	22.1 ± 2.1	—	—	—	—	—
323	19.0 ± 1.2	18.9 ± 1.4	28.2 ± 0.9 ^d	24.2 ± 2.1	25.5 ± 3.8	24.3 ± 1.9
333	—	—	—	—	23.0 ± 2.2	21.5 ± 0.9
338	—	—	22.7 ± 0.3	20.2 ± 0.7	—	—
348	16.8 ± 0.7	17.1 ± 1.2	21.3 ± 0.2	19.2 ± 0.6	21.8 ± 0.8	20.1 ± 0.7
[BMIM]PF ₆ (data: ref. 5)			[HMIM]PF ₆ (data: ref. 5)		[OMIM]PF ₆ (data: ref. 5)	
323	26.5 ± 1.2	24.4 ± 2.8	26.9 ± 4.4	27.8 ± 2.3	—	—
333	—	—	25.3 ± 2.1	25.1 ± 2.3	31.9 ± 4.5	—
343	23.5 ± 0.6	24.3 ± 1.9	23.9 ± 1.0	24.5 ± 0.8	26.0 ± 2.0	—
348	—	—	—	—	26.1 ± 1.3	31.8 ± 5.2
358	—	—	—	—	24.9 ± 1.2	29.1 ± 2.6
[Pyr ₁₄][Tf ₂ N] (data: ref. 2) ^e			[P _{222,5}][Tf ₂ N] (data: ref. 6) ^e			
303	30.5 ± 1.8	27.4 ± 1.5	—	—	—	—
313	—	—	29.4 ± 2.5	28.7 ± 2.1	—	—
323	26.0 ± 0.9	25.9 ± 1.0	28.1 ± 1.1	26.9 ± 1.3	—	—
338	24.6 ± 0.6	23.8 ± 0.7	23.8 ± 0.7	23.9 ± 0.7	—	—
348	23.0 ± 0.5	22.6 ± 0.6	23.1 ± 0.6	22.0 ± 0.7	—	—

^a $\Delta V^\ddagger$ units: cm³ mol⁻¹ calculated from data in the references cited. The $\ln D_{\text{Si}}$ isotherms are assumed linear in p for this purpose, though in fact the longer isotherms obtainable at higher temperatures are usually non-linear, as is common for liquid transport properties, making $\Delta V^\ddagger$ pressure dependent with $(\partial(\Delta V^\ddagger)/\partial p)_T < 0$ below 300–400 MPa (Fig. 6). The precision of the values in the table depends on the length of the isotherm. Note also that $(\partial(\Delta V^\ddagger)/\partial T)_p > 0$ in all cases. ^b Errors tabulated are the standard deviation from linear fits of $\ln[D_{\text{Si}}(p)]_T$ plus that due to the experimental uncertainty in D_{Si} . ^c The longer, concave isotherms at 338 and 348 K, are best fit to a binomial in p . This is consistent with the cubic dependence found for many liquids at sufficiently high pressure, particularly for the viscosity. ^d The larger difference between the cation and anion $\Delta V^\ddagger$ values for [HMIM]BF₄ is due to the short cation isotherm, 22.5 MPa in length. ^e [BMIM]⁺: 1-butyl-3-methylimidazolium; [HMIM]⁺: 1-hexyl-3-methylimidazolium; [OMIM]⁺: 1-octyl-3-methylimidazolium; [Pyr₁₄]⁺: *N*-butyl-*N*-methylpyrrolidinium; [P_{222,5}]⁺: triethylpentylphosphonium. [Pyr₁₄]⁺: *N*-butyl-*N*-methylpyrrolidinium; [P_{222,5}]⁺: triethylpentylphosphonium; [Tf₂N]⁻: bis(trifluoromethylsulfonyl)amide.

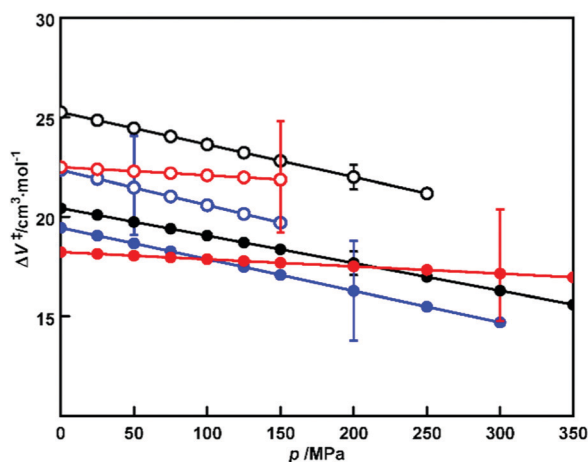


Fig. 6 $\Delta V^\ddagger$ as a function of temperature and pressure for the ion self-diffusion coefficients and the fluidity ($1/\eta$) of [EMIM][Tf₂N] (ref. 16), derived from the MVFT1 fits eqn (5). Symbols: open, 298 K; solid, 348 K; blue, cation self-diffusion; red, anion, self-diffusion; black, viscosity. The error estimates are larger than for the linear fits used in constructing Fig. 5 and Table 7 due to the influence of the uncertainty in the p^2 coefficient in eqn (5) (see Table 4).

pulses. Suarez *et al.* used fixed values of δ , Δ and τ , varying only g . We note that stimulated echo pulsed gradient methods^{49,50} are held superior to that employed by Suarez *et al.* and that Hayamizu has reported that self-diffusion measurements for ionic liquids at high viscosities (including [EMIM][Tf₂N]) show a dependence on the gradient pulse interval which must be varied to find a plateau region where the D_{Si} are constant.^{51,52} Values are anomalously high at short pulse intervals.

The apparatus used had two weak points. Firstly, there was no temperature control for the pressure vessel employed, so any increase in temperature caused by adiabatic compression when the pressure was adjusted may not have been adequately dissipated during experimental runs. We find it necessary to wait 10–15 min after pressure adjustments even with an efficient thermostat around the pressure vessel, which necessarily has a high heat capacity due to its mass. Secondly, the sample was contained within a plastic bag of ill-defined shape within an rf coil inside the pressure vessel. Consequently, the effective magnetic field gradient at the sample may not have been symmetric and may have been pressure dependent, depending on the compression and hence the shape of the sample volume.

This is important as the working equations assume right-cylindrical samples (as in an NMR tube; different equations apply for other geometries⁵³), with zero gradient along the plane bisecting the sample which must be correctly positioned at the centre of the field gradient. To meet these requirements, in our apparatus the sample is a right cylinder with a capillary connection to a thin Teflon bellows, whereby the sample can be compressed by the hydraulic fluid.⁵⁴ The radio frequency and quadrupolar magnetic field gradient coil lie within the pressure vessel, both wound on a ceramic former surrounding the cylindrical sample, positioned at the dead centre of the coils, and the bellows lies outside the range of their respective fields.^{27,55} A jig is used to move the pressure vessel to the centre of the field of the electromagnet as determined by the width of the unattenuated spin-echo.

From these considerations, it does seem that the apparatus and techniques employed by Suarez *et al.* were far from ideal for high-pressure self-diffusion measurements, and these may well have contributed to the differences observed between their results and ours.

Conclusions

High-pressure self-diffusion data are reported for 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide. The volumes of activation for the anion and cation self-diffusion coefficients do not differ significantly, in contradiction to the findings of Suarez *et al.*²⁰ The pressure dependence of the self-diffusion coefficients is similar to that for [BMIM][Tf₂N] and [HMIM][Tf₂N]; all three ionic liquids yield typical Stokes–Einstein–Sutherland diffusion–viscosity plots with overlaid isotherms and isobars collapsed onto a single line and with fractional exponents between 0.9 and 1, similar to other ionic and non-associated molecular liquids, whereas the Suarez data are inconsistent, with different slopes for the high-pressure isotherms and atmospheric pressure isobars; finally, the data conform to thermodynamic or density scaling, with very similar scaling constants for the self-diffusion coefficients and the viscosity, consistent with data for other ionic liquids and with a magnitude consistent with the theoretical predictions of Knudsen *et al.*⁴¹

In summary, it is found that there is nothing different in the effect of pressure on the self-diffusion of the ions in [EMIM][Tf₂N] to support a “slithering” mechanism for the [Tf₂N][−] ([TFSA][−]) ion in this ionic liquid distinct from the effects of compression on the translation of other cations or anions or in [Tf₂N][−] salts with larger 1-alkyl-3-methylimidazolium cations.

Author contributions

K. R. Harris: conceptualization, formal analysis, investigation (self-diffusion), resources, methodology, validation, writing – original draft, writing – review & editing, visualisation. M. Kanakubo: conceptualization, writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

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